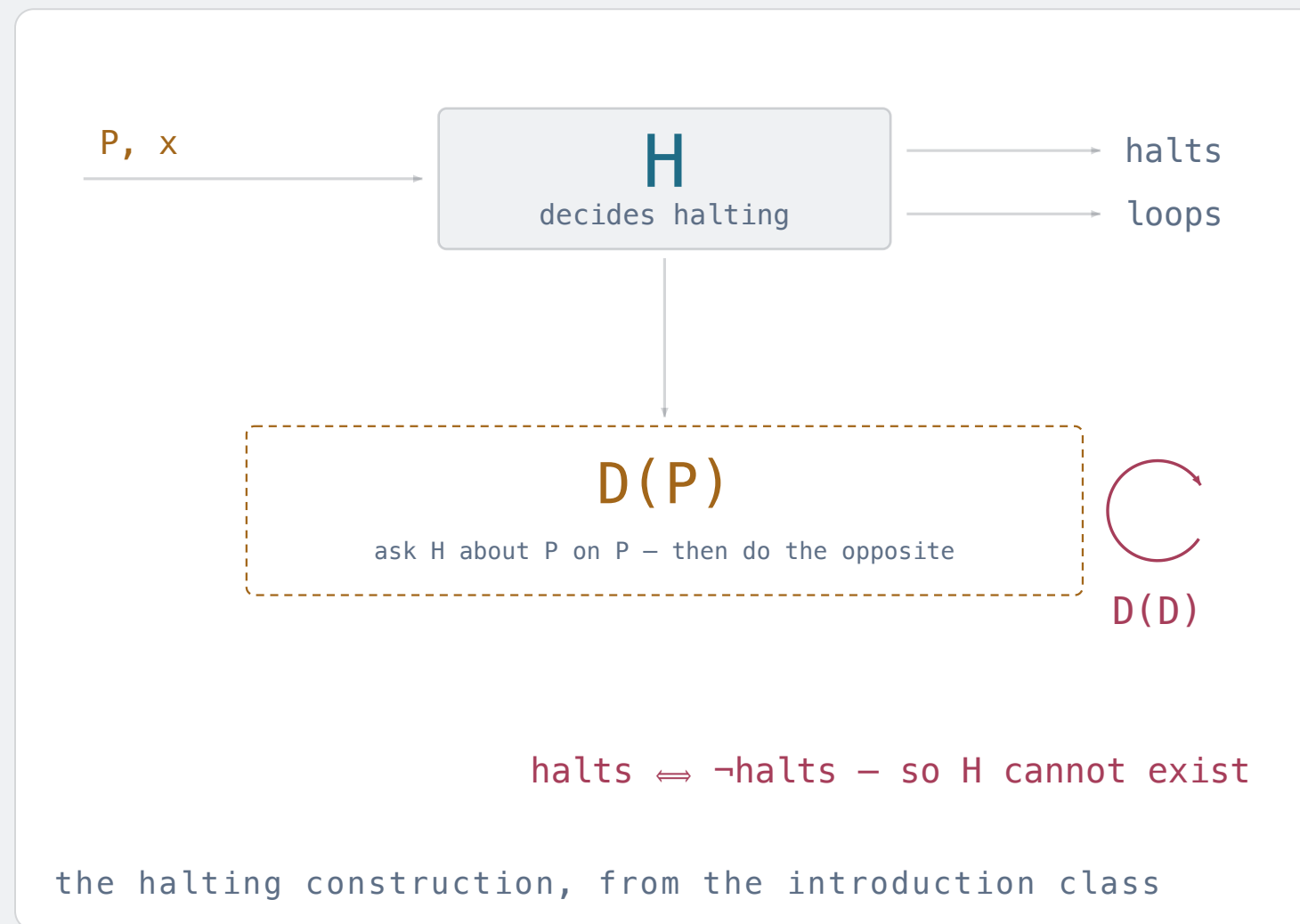


Time-Sharing

The timer interrupt is the kernel's answer to the halting problem.
Scheduling is what it cannot answer.

← SE 04 · Virtual Memory

Will this program give the processor back? **Undecidable**. So take it back.



Cooperative multitasking asks each program to yield. A program that never does takes the machine with it, and no kernel can screen for that in advance: Turing, 1936.

Preemptive multitasking does not ask. A hardware timer raises an exception after a fixed number of cycles; the kernel saves the context and runs another. The halting problem is not solved; it is made irrelevant by a bound.

Every kernel principle in this class has this shape: what cannot be decided is bounded.

A timeout is a number of instructions. Mipster counts down and **returns**.

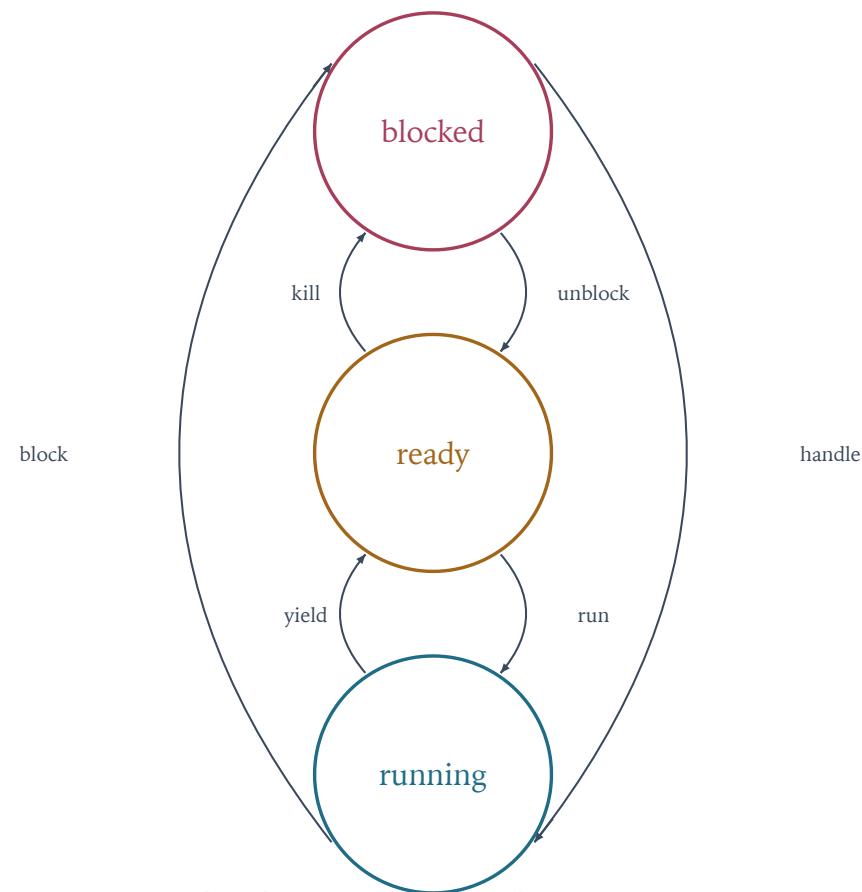
```
// selfie.c, the shape of it
uint64_t* mipster(uint64_t* to_context,
                  uint64_t timeout) {
    restore_context(to_context);
    run_until_exception(timeout); // TIMER is one
    save_context(to_context);
    return to_context;           // caller decides
}
```

The emulator's loop from week 3 with one more exit: after timeout instructions it raises a timer exception and returns the context, as it would on a system call or a page fault. The caller, the kernel, decides what happens next.

The kernel in selfie is a loop around this call: pick a context, run it for a slice, handle what stopped it, pick again. That loop is the operating system by emulation.

```
$ make os-emu
./selfie:           166 syscalls, 529 page faults, 1898 timer interrupts
```

Running, ready, blocked. A machine context is always in exactly one.



the three states and the events that move a context between them

Running: its registers are in the processor. Ready: saved, waiting for a slice. Blocked: saved, waiting for input, a child, a lock. The timer moves running to ready; a system call that must wait moves running to blocked; the awaited event moves blocked to ready.

A finite state machine per context. The kernel's whole knowledge of a program is which of three states it is in and a few kilobytes of saved registers. By Rice it knows nothing about what the program is doing, and needs to know nothing.

Which ready context runs next? Fair is easy. Optimal is NP-hard.

ROUND ROBIN

A queue; the timer moves the running context to the back. Every context gets a slice every n slices. Fair, simple, and what selfie does. It optimises nothing.

PRIORITIES, DEADLINES

Real schedulers weigh interactivity, deadlines, fairness across users, energy. Each is a policy; each can be gamed; each is a bound on something that cannot be decided.

THE LIMIT

With deadlines and several processors, finding a schedule that meets every deadline is NP-hard: one more job multiplies the schedules to try. Even knowing every job's length, which by Rice it cannot, the kernel could not afford the optimum.

Two limits, one design: the scheduler does not know and could not compute. So it is a policy, chosen by people, adjusted by measurement. Week 11 measures.

ASSIGNMENT

processes: run several instances of the same program, time-shared.

- 1 Extend selfie so that `-m` can run n instances of the loaded program as separate machine contexts with separate memory, round-robin scheduled by the kernel loop with a timeout you choose.
- 2 Each instance prints something that identifies it, so that the interleaving is visible on the terminal. Make the slice small enough that the interleaving shows.
- 3 Check that no instance can see another's memory, and that the total instruction count is the sum of the instances'.
- 4 `./grader/self.py processes`.

REUSE

`create_context`, `load`, the kernel loop, and `mipster` with a timeout are all there. The assignment is to arrange them into a list of contexts and a scheduler. Look at how the existing code handles a context that exits, and keep the case of one instance behaving exactly as before.

Listen, then read.

LISTEN

Bach · Wachet auf, ruft uns die Stimme — Netherlands Bach Society. The watchman's call at midnight, and the sleepers who did not budget their oil: the timer interrupt, with consequences. Time-sharing has a cantata.

READ

An Experimental Time-Sharing System — Corbató, Merwin-Daggett and Daley, 1962: CTSS at MIT, the first time-sharing system, with the timer interrupt as its centre. And, more technical, Scheduling Algorithms for Multiprogramming in a Hard-Real-Time Environment — Liu and Layland, 1973: the scheduling theory, rate-monotonic and earliest-deadline-first, with the proofs.